BEHAVIORAL BIOMETRIC AUTHENTICATION SYSTEM

Mouse Dynamics · Keystroke Analysis · Touchscreen Gestures

P003 Computer & Network Security · 2025–2026

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33.19%
Mouse EER

12.87%
Keystroke EER

28.60%
Touchscreen EER

718
Users Tested

OVERVIEW & MOTIVATION

"Can the natural, unconscious interaction patterns of a user — typing rhythm, mouse kinematics, and gesture dynamics — serve as a reliable, continuous authentication factor capable of detecting impostors even when they hold valid credentials?"

🔑 The Problem

- Passwords can be stolen, guessed or phished
- No verification after login
- Credentials ≠ Identity

👁 The Insight

- Every keystroke carries a behavioral fingerprint
- Motor patterns are personal & subconscious
- Shaped by years of individual learning

✓ Our Solution

- Continuous authentication via 3 modalities
- ML anomaly detection per user
- Combined score threshold ≥ 0.60

SYSTEM ARCHITECTURE

01 · DATA LAYER

Browser JS event listeners capture mousemove, keydown/keyup, touchstart/move/end

→ Timestamped streams to Flask via HTTP POST

02 · ML LAYER

Feature extraction into fixed-length vectors

Per-user model storage (OC-SVM, IF, RF)

Normalized authentication scores [0,1]

03 · INTERFACE LAYER

Four-tab web app:

Enroll · Authenticate · Statistics · How It Works

Real-time score histograms & decisions

COMBINED AUTHENTICATION LOGIC

Keystroke Dynamics Score

Weight: 50% · Threshold: ≥ 0.60

Both must pass simultaneously — neither score may compensate for the other

Mouse Dynamics Score

Weight: 50% · Threshold: ≥ 0.60

Both must pass simultaneously — neither score may compensate for the other

MACHINE LEARNING MODELS

Isolation Forest

MOUSE & TOUCH

Ensemble anomaly detector using random binary partitions.

Anomaly score = $2^{(-E[h(x)]/c(n))}$

Short path → anomaly; long path → genuine

Config: 100–200 trees, contamination 0.05–0.1

✓ *Fast, robust to irrelevant features*

One-Class SVM

TOUCH (baseline)

Fits a hypersphere in kernel (RBF) space around genuine data.

Classifies genuine if $\mathrm{sign}(w \cdot \Phi(x) - \rho) > 0$

nu=0.1 bounds the fraction of outliers

Ideal for nonlinear, non-convex behavioral clusters

✓ *Explicit boundary, interpretable*

Per-User Random Forest

KEYSTROKE (primary)

Binary classifier: genuine vs. impostors (1:3 ratio)

100 trees, max depth 10, class_weight=balanced

Top features: dwell_mean, dwell_std, dd_mean

Decision threshold calibrated at 0.60

✓ *Best EER — learns explicit boundaries*

PART I — MOUSE DYNAMICS

21-DIMENSIONAL FEATURE VECTOR

Velocity

`speed_mean, speed_std, speed_25, speed_75`

Acceleration

`accel_mean, accel_std, accel_25, accel_75`

Curvature

`angle_mean, angle_std, angle_25, angle_75`

Spatial

`x_range, y_range, total_distance, avg_speed`

Temporal

`session_duration, avg_pause, long_pauses, click_count, move_count`

RESULTS & ANALYSIS

Dataset

14 users · 196 sessions · EVTRACKTRACK.csv
Feb 2020 – Jan 2024 · 143,115 records

Model

Isolation Forest (n=100, contamination=0.05–0.10)
Threshold = 10th percentile of training scores

Why EER is high (33.19%)

Small cohort (14 users) → behavioral overlap in feature space
OC learning cannot see where one user's cluster ends
Some users have only 3 training sessions (memorizes ≠ generalizes)

Live Registration Improvement

Same-day enrollment reduces within-session variance
Tighter model boundary → better genuine/impostor separation

PART II — KEYSTROKE DYNAMICS

TIMING FEATURES

DWELL RELEASE − PRESS · Mean: 109ms

→ *most person-specific*

FLIGHT PRESS[k+1] − RELEASE[k] · Mean: 116ms

→ *expert typists show negatives*

DD PRESS[k+1] − PRESS[k] · Mean: 225ms

→ *most cited in literature*

SLIDING WINDOW

Window: 20 keystrokes · Step: 10 (50% overlap)

21 statistical features per window: mean, std, median, IQR, skew for each timing column

Train/test split at session level (no data leakage)

MODEL COMPARISON RESULTS

Model	Dataset	EER	AUC
OC-SVM	Aalto	18.14%	0.886
RF (primary)	Aalto	**12.87%**	0.929
OC-SVM	KeyRecs	25.95%	—
RF	KeyRecs	18.05%	—

RF outperforms OC-SVM by 5.3pp on Aalto (718 users)

because it learns explicit user-vs-impostor boundaries.

Enrollment depth effect: +13 sessions = −5.2pp EER.

PART III — TOUCHSCREEN GESTURE DYNAMICS

PER-USER RESULTS (5 USERS)

User	IF EER%	IF AUC	SVM EER%	SVM AUC	Best
0	25.00	0.700	60.00	0.560	IF
1	33.33	0.757	27.08	0.830	SVM
2	18.52	0.883	25.93	0.827	IF
3	50.00	0.680	30.00	0.640	SVM
4 ★	**16.67**	**0.806**	**0.00**	**1.000**	**SVM ★**
Mean	**28.70**	0.765	**28.60**	0.771	—

★ User 4: Perfect authentication — EER = 0%, AUC = 1.00
Pressure profile + direction_change wholly non-overlapping

19-DIM FEATURE VECTOR

Temporal

`duration, num_events`

Spatial

`start/end (x,y), length, length_x/y`

Kinematic

`speed_mean/std, accel_mean/std`

Pressure ★

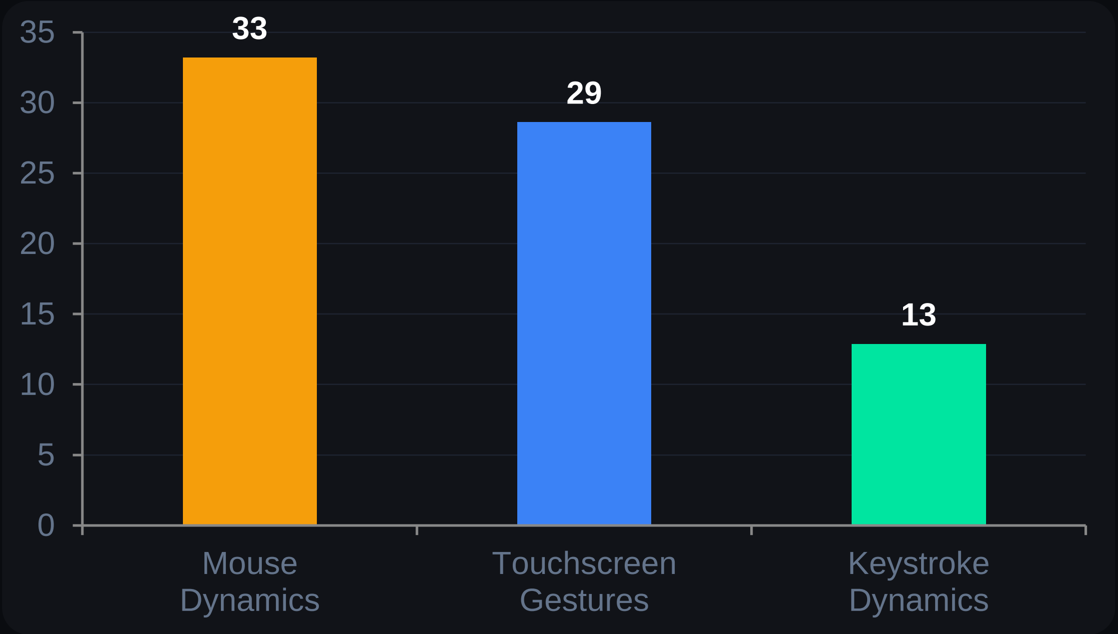
`pressure_mean/std, pressure_25/75`

Direction

`direction_angle, direction_change`

Pressure is unavailable in mouse tracking — this extra dimension helps differentiate touch users more distinctly.

CROSS-MODALITY RESULTS SUMMARY



WHY KEYSTROKE WINS

Random Forest benefits from explicit impostor contrast — a structural advantage OC approaches cannot replicate.

Touch beats Mouse (28.6% vs 33.19%) because pressure features are unavailable in mouse tracking — richer 19D space improves separation.

More enrollment sessions directly lower EER: +13 sessions = −5.2pp.

Modality	Algorithm	Mean EER	Best User	N Users
Mouse Dynamics	Isolation Forest	33.19%	—	14
Keystroke	Per-User RF	12.87%	<10% for 271/717	718
Touchscreen	One-Class SVM	28.60%	0% EER (User 4)	5

REAL-TIME AUTHENTICATION SYSTEM — LIVE DEMO

🔗 https://spontaneous-brioche-76fba3.netlify.app/

ENROLL

User types freely for 2–3 min (≥200 keystrokes) and records 3 mouse sessions.
Backend trains personal model in ~2–3 seconds.

AUTHENTICATE

Real-time color-coded histogram per 20-keystroke window.
Green = ACCEPT, Red = ALERT. Moving average of last 5 windows.

STATISTICS

Per-user FAR, FRR, EER charts.
Feature importance view.
Live anomaly score timeline.

HOW IT WORKS

Visual walkthrough of the ML pipeline and feature extraction for each modality.

⚡ Impostor detected within 60–90 seconds (3–5 scored windows) when a different user takes over the keyboard

LIMITATIONS & FUTURE WORK

Mouse EER 33.19%

Cause: Small cohort (14 users), behavioral overlap

→ *Scale to 50+ users; add click dynamics features*

Static decision thresholds

Cause: No adaptation to behavioral drift over time

→ *Adaptive per-session threshold recalibration*

Touchscreen EER 28.6%

Cause: Only 5 users; 232 gestures

→ *Larger cohort; convolutional gesture encoders*

Lab conditions only

Cause: No fatigue, stress, or device variation

→ *Longitudinal field study; multi-device testing*

Keystroke EER 12.87% vs SOTA 3–8%

Cause: ~43 windows/user on Aalto dataset

→ *Longer enrollment; Siamese network session similarity*

OC-SVM $O(n^2)$ training

Cause: No impostor data for mouse/touch enrollment

→ *Semi-supervised approach with shared impostor pool*

CONCLUSION

KEY FINDINGS

Keystroke (12.87% EER) — strongest result using supervised per-user RF with dwell time as #1 feature.

Touchscreen (28.6% EER) — pressure-augmented 19D features; User 4 achieved perfect 0% EER / AUC 1.0.

Mouse (33.19% EER) — limited by small cohort (14 users); significant improvement expected with scale.

SECURITY ARGUMENT

The multi-modal architecture creates an authentication surface that goes beyond static credentials. Even if an attacker obtains a user's password, they must still replicate the enrolled user's typing rhythm AND mouse kinematic profile simultaneously — a task requiring deep behavioral mimicry, not just knowledge.

12.87% Best EER (Keystroke RF)

718 Users Keystroke

0% User 4 EER Touchscreen

~60s Impostor Detection Time